

Appendix 1: FKB327-001 protocol outline

Study Protocol

Study title	A randomized, double-blind, single-dose study to compare pharmacokinetic characteristics and safety of FKB327 with those of Humira® in healthy subjects
Short title	Comparing the pharmacokinetics and safety of FKB327 and Humira®
Sponsor code	FKB327-001
Study medication	FKB327 and Humira®
Phase of study	Phase 1

Summary

1.1 Study medication

FKB327 is being developed as a similar biological medicinal product to Humira[®]. It is a humanized monoclonal antibody that binds selectively to tumour necrosis factor alpha (TNF- α).

1.2 Objectives

Primary: To compare the safety and pharmacokinetics of FKB327 and EU- and USA-licensed Humira[®] after single doses, by subcutaneous (SC) injection in healthy volunteers

Secondary: To assess immunogenicity and tolerability after single doses of FKB327 and EU- and USA-licensed Humira[®], by SC injection

1.3 Type of study

Phase 1, randomised, double-blind, parallel group, single dose study in healthy volunteers.

1.4 Study population

a **Total** 180

b **Inclusion criteria**

Male volunteers, and female volunteers of non-childbearing potential, aged 18–65 years, deemed healthy on the basis of medical history, physical examination, electrocardiogram (ECG), vital signs and laboratory tests of blood and urine; weight 60–90 kg; body mass index of 18–30 kg/m².

c **Exclusion criteria**

Pregnant or lactating women; clinically relevant abnormal medical history or abnormal findings at the screening assessment; acute or chronic illness or history of chronic illness; clinically relevant concurrent medical condition; history of cancer; severe adverse reaction or history of anti-drug antibodies (ADAs) to any drug; previous dosing with adalimumab; evidence of any significant bacterial, viral, fungal or parasitic infection during the 4 weeks before dosing; minor infection not requiring anti-infective treatment during the 2 weeks before dosing; active or latent tuberculosis; loss of more than 400 mL blood, within the previous 3 months; objection by General Practitioner to the volunteer entering study.

1.5 Study design and methods

This is a parallel group study. 180 subjects will be randomly assigned to receive a single 40 mg SC injection of EU-licensed Humira[®], USA-licensed Humira[®] or FKB327, in a 1:1:1 ratio.

Subjects will be dosed on the morning of Day 1. Subjects will remain resident in the ward from Day –1 (the morning before Day 1) until 192 h after their dose (Day 9). They will return for outpatient visits on Days 16, 23, 30, 37, 44, 51 and 65.

The following assessments will be made:

Safety: laboratory assessments (routine haematology, biochemistry and urinalysis), physical examination, 12-lead ECG, telemetry, vital signs, and adverse events.

Tolerability: local tolerability (including injection site reactions and pain) and adverse events.

Pharmacokinetic: blood samples for assay of serum adalimumab concentration will be taken before, and periodically up to 1536 h (Day 65) after dosing.

Immunology: blood samples for detection of ADAs will be taken before, and at 360 h (Day 16), 696 h (Day 30) and 1536 h (Day 65) after dosing.

1.6 Primary variables

Safety: vital signs, ECG, physical examination, laboratory safety tests

Tolerability: adverse events

Pharmacokinetics: C_{max} , $AUC_{0-\infty}$ and AUC_{0-t} of adalimumab

1.7 Secondary variables

Tolerability: local site reaction assessments and visual analogue scales (VAS) for pain

Pharmacokinetics: t_{max} , AUC_{0-360h} and $t_{1/2}$ of adalimumab (additional parameters may be calculated if deemed necessary)

Immunology: presence of ADAs

Study flow chart

	Screening	-1	Study day hours post-dose																					
			1						2		3	4	5	6	7	8	9	16	23	30	37	44	51	65
			pre-dose	0	0.5	1	4	12	24	36	48	72	96	120	144	168	192	360	528	696	864	1032	1200	1536
Informed consent	X																							
Medical history/ demographics	X																							
Physical examination, Vital signs, ECG, Telemetry, Laboratory testing are assessed at the schedule defined by protocol	X	X																						
Dosing				X																				
Local tolerability				X ¹				X	X		X	X	X				X							
VAS				X ¹	X	X		X	X															
Blood for serum adalimumab PK			X				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Blood for serum ADA			X														X		X					X

1. Immediately after dosing.